1. Product Overview

WIN-SN-7-1Enviro-M is a high-performance, multi-parameter (7 in 1) environmental sensing module engineered for accurate and reliable air quality monitoring across industrial, commercial, and IoT applications. The device integrates advanced sensing elements to simultaneously measure CO_2, CO, NO_2, O_2 within a single compact enclosure.

The module incorporates a CO_2 sensing element (400 – 5000 ppm range), CO sensing element (0 – 500 / 0 – 1000 ppm range), O_2 sensing element (0 - 30% range), NO_2 sensing element (0 – 10 / 0 – 20 ppm range). All sensor outputs are internally processed, calibrated, and digitally compensated to ensure high measurement accuracy and long-term stability.

Fig.1 - Product Image

Communication is facilitated through a robust RS485 interface using the Modbus RTU protocol, enabling seamless integration with PLCs, SCADA systems, building management systems (BMS), and embedded controllers.

The device supports reliable long-distance communication and noise immunity required for industrial environments.

Designed for continuous operation, the sensor module features low power consumption, compact form factor, and rugged construction, making it suitable for both indoor and semi-outdoor deployments. Its durable design ensures stable performance under varying environmental conditions, including exposure to dust, airflow disturbances, and extended operational cycles.

This solution provides a cost-effective and scalable platform for real-time air quality monitoring in applications such as smart buildings, HVAC control systems, environmental monitoring stations, and industrial automation systems.

2. Key Features:

- **Integrated Multi-Sensor Platform**
  - CO_2, CO, NO_2, O_2
- **High Measurement Accuracy**
  - Factory-calibrated sensors with digital compensation for reliable and consistent reading
- **Industrial Communication Interface**
  - RS485 interface with **Modbus RTU protocol** for seamless integration with industrial systems
- **Wide Measurement Capabilities**
  - CO_2, CO, NO_2, O_2
- **Fast Response and Real-Time Monitoring**
  - Optimized sensor fusion for rapid detection of environmental changes
- **Robust and Compact Design**
  - Space-efficient enclosure suitable for embedded and field installations
- **Wide Operating Conditions**
  - Temperature Range: **-10°C to 60°C**
  - Humidity Range: **0% to 80% RH (non-condensing)**
- **Low Power Consumption**
  - Suitable for IoT and continuous monitoring applications
- **Long-Term Stability**
  - Designed for extended operation with minimal drift
- **Ease of Integration**
  - Pre-calibrated digital output reduces system-level complexity

Sensor specification

Measuring Parameter	Measuring Range
O_2	0 - 30%
CO_2	400 – 5000 ppm
CO	0 – 500 / 0 – 1000 ppm
NO_2	0 – 10 / 0 – 20 ppm

3. Technical Specifications:

- **Operating Voltage:** 12-24v DC
- **Interface:** RS485
- **Function Code:** 0x03
- **Product dimension:** 85*85*40 mm
- **Certifications:**

4. Configuration Settings

Communication Speed	Software Configurable **(9600, 19200, 38400, 57600, 115200)**
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	Software Configurable **(1-247)**
Function code	0X03 (Read Holding Register)

Recommended Cable Electrical Characteristics

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

1. Modbus Register Values

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	O2	40000	0x03	RS485	16-bit int
2	CO2	40001	0x03	RS485	16-bit int
3	CO	40002	0x03	RS485	16-bit int
4	NO2	40003	0xo3	RS485	16-bit int

2. Modbus RS485 Data Configuration register

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	Display Baud Rate (Default: 960)	40010	0x03	RS485	16-bit int
2	Enter New Baud Rate	40011	0x03	RS485	16-bit int
3	Display Slave ID (Default: 1)	40012	0x03	RS485	16-bit int
4	Enter New Slave ID	40013	0x03	RS485	16-bit int
5	Display Parity (Default: None-3 And for Odd-1, Even-2)	40014	0x03	RS485	16-bit int
6	Enter New Parity	40015	0x03	RS485	16-bit int
7	Display Stop Bit (start-2/stop bit-1)	40016	0x03	RS485	16-bit int
8	Enter New Start/ Stop Bit	40017	0x03	RS485	16-bit int

7. Important Note

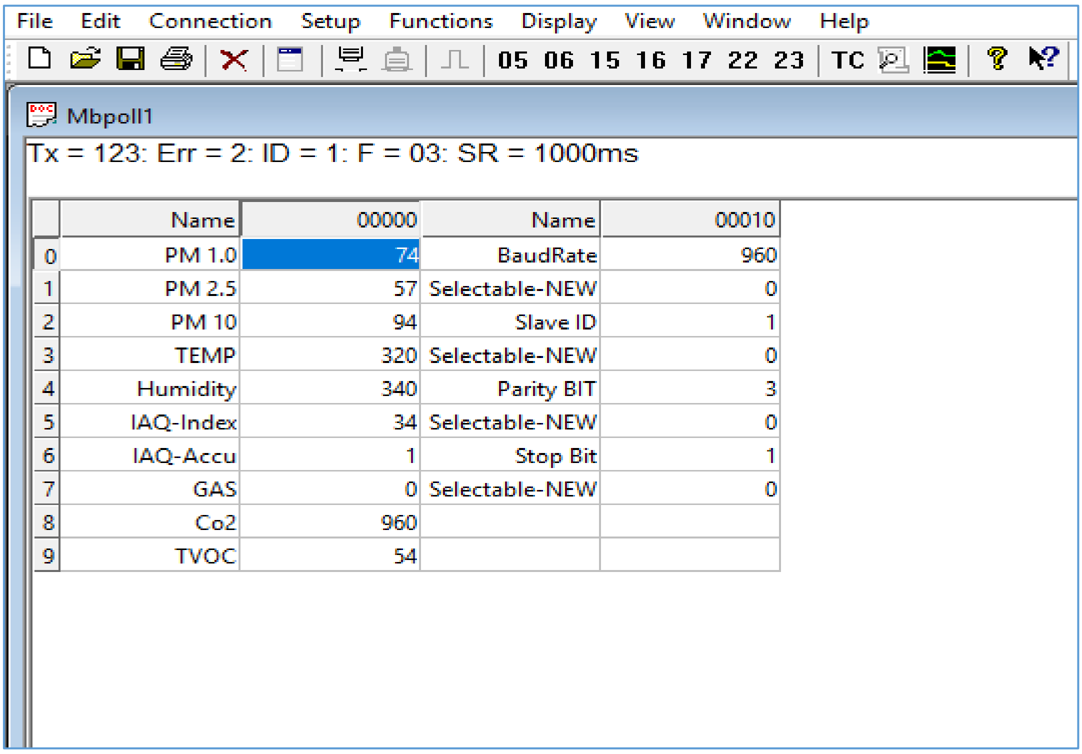
1. If you want to enter the baud rate 9600/19200/38400/57600/115200etc, enter it like this 960/1920/3840/5760/11520etc.
2. After that press the reset button or reboot the system.
3. Then you will get the present value of 11520 at 40010 baud rate (115200) and 10 at 40012 slave id.
4. PM1.0, PM 2.5, PM 10 in ug/m3 Data are provided in the multiple of 10, for the sake of higher resolution, you need to divide it by 10 to obtain the actual reading. Example. PM 1.0 = 912/100 = 91.2 ug/m3.
5. Baud rate is provided in devised of 10. Example. Baud rate 960*10=9600.

8. Modbus Configuration

Here is an example of receiving data on Mb poll software.

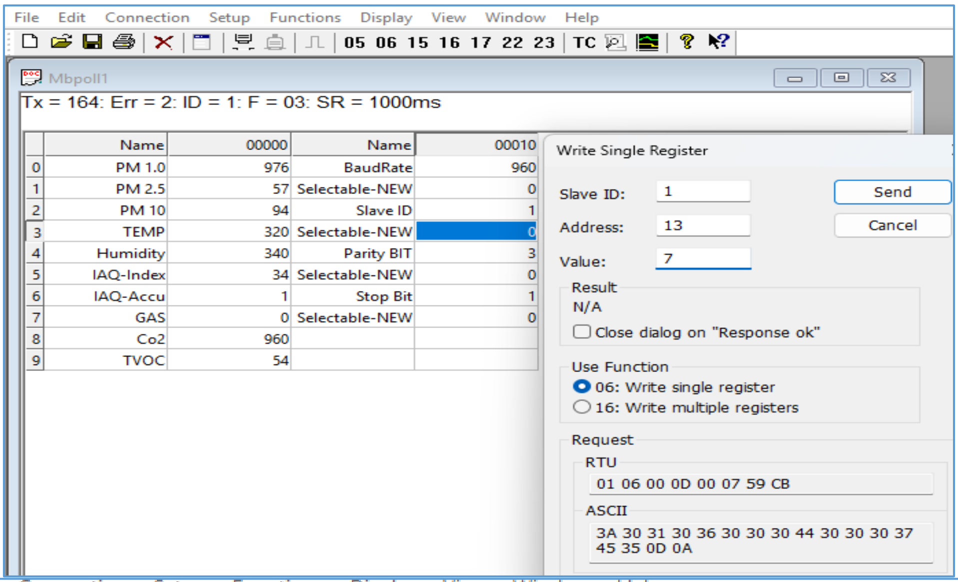
- Open Mb-poll software connect using configuration setting (Default baud rate is 9600 and slave id is 1).

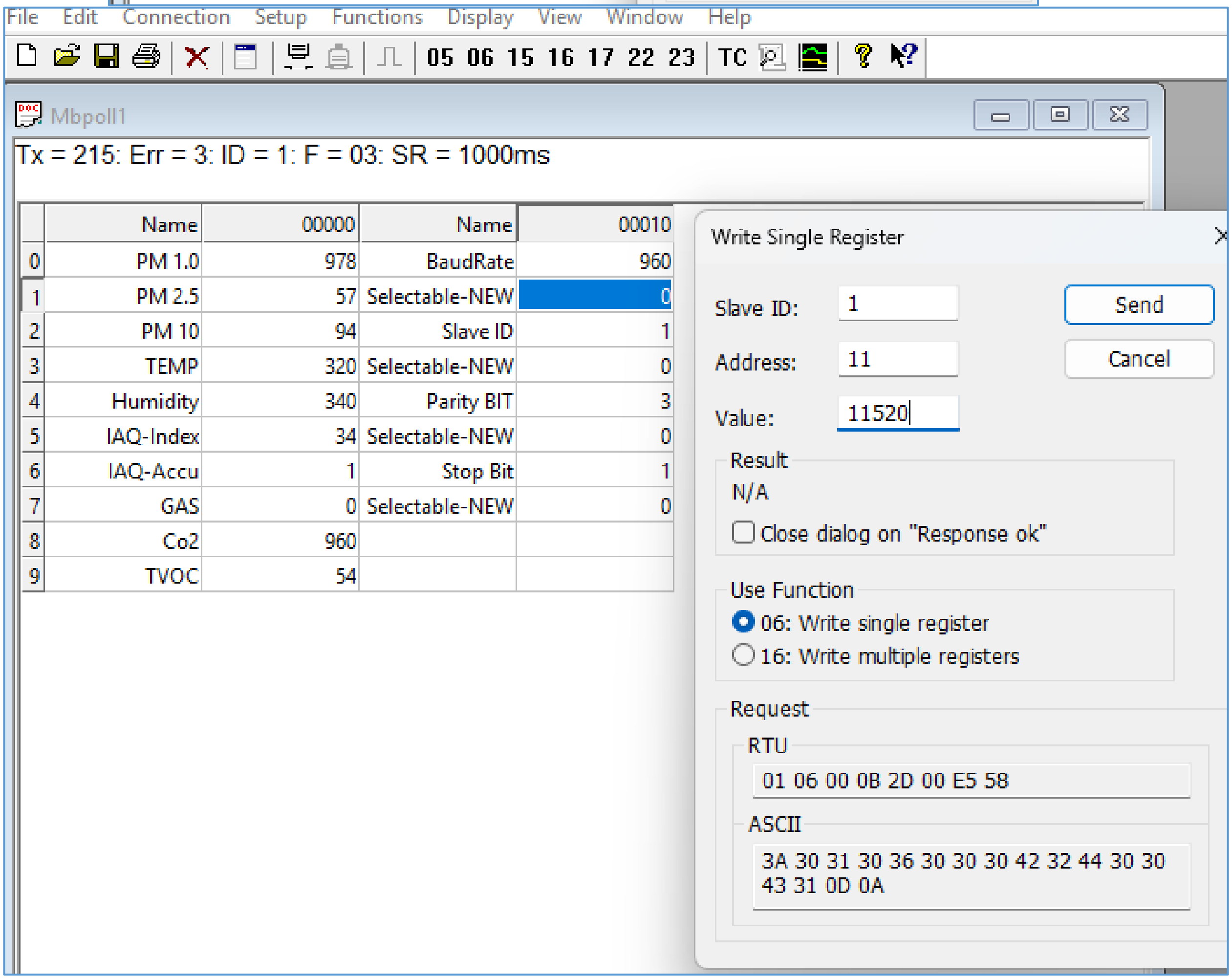
- After connecting you will get data as shown below



	Name	00000	Name	00010
0	PM 1.0	74	BaudRate	960
1	PM 2.5	57	Selectable-NEW	0
2	PM 10	94	Slave ID	1
3	TEMP	320	Selectable-NEW	0
4	Humidity	340	Parity BIT	3
5	IAQ-Index	34	Selectable-NEW	0
6	IAQ-Accu	1	Stop Bit	1
7	GAS	0	Selectable-NEW	0
8	Co2	960		
9	TVOC	54		

- As you can see in the image above, the resistor 40010 – 960 represents the present bund rate (9600) of the board and 40012 – 1 represents the present slave id of the board.
- To change the baud rate and slave id you must enter a value on the 40011 resistors for the baud rate and 40013 for the slave id as shown in the image below.
- Now you can see data. In below image set 115200 baud rate and slave ID 10.





Hard reset Setting

Somehow you forgot the baud rate or slave id of the board then remove the jumper (as shown in the image) from the board reboot the system or press the reset button (given on the board). Then you will get the default baud rate of 9600 and slave id 1.

9. Installation & Wiring

- Ensure correct power supply (12 V DC or 230 V AC version).
- Mount probe in the location representative of the environment (airflow, avoid direct sunlight or heating elements).
- Ensure the enclosure is mounted such that the probe can access the air freely (not blocked).
- Use **software interface** to configure baud rate, slave ID, parity, stop bits, etc.
- For proper RS485 network: Use **twisted-pair, shielded cable;** keep line length within specification; include termination resistor if required.
- Ambient conditions: Keep the probe and electronics within their specified operating temperature and humidity ranges.
- Do not open the enclosure or modify wiring while powered.

10. Precautions & Maintenance

- Do **not** disassemble or modify the sensor board – this may void **warranty and affect calibration.**
- Keep away from **dripping water**, condensation, high humidity above specification, corrosive gases.
- Ensure airflow around probe sensor is not obstructed to guarantee accurate response.
- Use only the **specified power supply;** reverse polarity or over-voltage may damage the unit.

- Periodically verify calibration of temperature/humidity sensors especially in critical applications.

Contact Information

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